The Equation and Lewis's triviality results

Topics in Language and Mind: Conditionals

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1. Unfinished business

(Incremental) Bayesianism: if S has credence function Cr at t and is (ideally) rational throughout the interval from t to t+, then for some proposition E, S's credence function at t+ is Cr_E. ('Credences evolve by conditionalization').

- What proposition should one conditionalize on? Bayesians say: "on one's total evidence". But 'evidence' here is used in a somewhat technical sense.
- Probabilists who reject Incremental Bayesianism want to allow for (i) rational forgetting, and/or (ii) evolution of credence without becoming absolutely certain of anything. 'Bayesian' is often used loosely in a way that covers some alternatives to Incremental Bayesianism.

2. The 'Ramsey Test'

> If two people are arguing 'If A will C?' and are both in doubt as to A, they are adding A hypothetically to their stock of knowledge and arguing on that basis about C. . . . We can say they are fixing their degrees of belief in C given A. (Ramsey 1929: 143)

This suggestive remark can be taken in various directions. For example:

Adams's Thesis: the 'degree of assertability' of an indicative conditional (for a speaker at a time) equals the speaker's conditional degree of belief in its consequent given its antecedent.

- What is a "degree of assertability"? I don't know.

3. The Equation

> **Schematic Credence Equation:** Necessarily, at any time, any (ideally) rational person's credence that if P, Q equals their conditional credence in Q given P, provided their credence in P is positive.

4. The Chance Equation

> **Schematic Chance Equation:** Necessarily, at any time, the chance that if it were the case that P it would be the case Q equals the conditional chance of Q given P, provided the chance of P at that time is positive.

- For example: if the proposition that the coin was tossed at 12.01 had a chance of 0.2 at noon, and the proposition that the coin was tossed and landed Heads had a chance of 0.1 at noon, the conditional chance at noon of the latter given the former was 0.5; so according to the Chance Equation, the proposition that the coin would have landed Heads if it had been tossed had a chance of 0.5 at noon.

5. Controlling for context sensitivity

Our formulations of the Equation and the Chance Equation are not ideally clear, given that we have seen some good reasons to take seriously the idea that 'if' sentences are systematically context sensitive.

- On the strong interpretations of the Equations, there is a *single* reading of 'if P, Q' or 'if it were the case that P it would be the case that Q' under which they hold for *every* time (and in the case of the Credence Equation, for every person). But we can also consider a weaker interpretation, where we are allowed to associate different resolutions of the context sensitivity of the conditionals with different times (or different $\langle$person, time$\rangle$ pairs).
- For example, in the case of counterfactuals, we considered a picture where for every time, there is a distinct interpretation of 'If P it would be that Q' where the accessible worlds are those that match history up to that time. Given this semantics, we might expect that the relevant interpretation when we are applying the Chance Equation to t is the one where accessibility is match up to t.
- In the case of indicatives, we considered a picture where for every person and time, there is a distinct interpretation of 'If P, Q' where the accessible worlds are those compatible with that person's knowledge/evidence at that time. Given this semantics we might expect that the relevant interpretation when we are applying the Credence Equation to S at t is the one where accessibility is compatibility with what S knows at t.

To help us avoid running together strong and weak interpretations, let's introduce some definitions. Here, $\Rightarrow$ is any binary operation (function from pairs of propositions to propositions); Pr is any probability function; and **R** is any set of probability functions.

$\Rightarrow$ **is equational for Pr** iff whenever Pr(A)>0 and Pr(B)≥0, $\Pr(A{\Rightarrow}B) = \Pr(B \mid A)$.

$\Rightarrow$ **is equational for R** iff $\Rightarrow$ is equational for every $\Pr \in R$.

Now we can distinguish:

> **Invariantist Chance Equation:** Some binary operation $\Rightarrow$ is equational for the set of all probability functions that could be the chance function at some time. The counterfactual 'if' has a reading where it expresses such a $\Rightarrow$.

> **Invariantist Credence Equation:** Some binary operation $\Rightarrow$ is equational for the set of all probability functions that could be the credence function of a ideally rational person at some time. The indicative 'if' has a reading where it expresses such a $\Rightarrow$.

And on the other hand:

Contextualist Chance Equation: In a context where *t* is salient, the counterfactual 'if' is naturally interpreted as expressing a binary operation that is [necessarily?] equational for the chance function at *t*.

Contextualist Credence Equation: In a context where person *S* and time *t* are salient (e.g. when S is the speaker and *t* is the time of speech), the indicative 'if' is naturally interpreted as expressing a binary operation that is [necessarily?] equational for S's credence function at *t* if S is rational at *t*.

6. Lewis-style triviality results

Lewisian Lemma: Suppose Pr is a probability function and A and B are propositions such that A entails B and $0<\Pr(A)<\Pr(B)<1$, and $\Rightarrow$ is a binary operation that is equational for Pr. Then $\Rightarrow$ is not also equational for $\Pr_B$.

- *Consequence:* no binary operation is equational for a set of probability functions such that some member of the set is both derived from another by conditionalisation and takes values other than 0 and 1.
- Given that rational credences *can* evolve by conditionalisation (an assumption much weaker than Incremental Bayesianism) this refutes the Invariantist Credence Equation.
- Given that chances evolve by conditionalization on history, this refutes the Invariantist Chance Equation.

Proof of Lewisian Lemma:

1. Given Pr, A, B, and $\Rightarrow$ as described: let C be $A \vee \neg B$.
2. Note that $0<\Pr(C)<1$ and $0<\Pr_B(C)<1$.
3. Since $\Pr(C)>0$ and $\Rightarrow$ is equational for Pr, $\Pr(C \Rightarrow \neg B) = \Pr(\neg B \mid C) = \Pr(\neg B \wedge C)/\Pr(C)$ $= \Pr(\neg B)/\Pr(C) > \Pr(\neg B)$.
4. Hence $\Pr(B \wedge (C \Rightarrow \neg B)) > 0$.
5. So, $\Pr_B(C \Rightarrow \neg B) = \Pr(B \wedge (C \Rightarrow \neg B))/\Pr(B) > 0$.
6. But $\Pr_B(\neg B \wedge C) = \Pr(\neg B \wedge C \wedge B)/\Pr(B) = 0$.
7. Hence $\Pr_B(\neg B \mid C) = \Pr_B(\neg B \wedge C)/\Pr_B(C) = 0$.
8. So by lines 4 and 6, $\Rightarrow$ is not equational for $\Pr_B$.

7. Lewis's first three theorems

They are much weaker (and thus less telling against the Invariant Equations) than our Lemma, but here goes:

First triviality theorem: no binary operation $\Rightarrow$ is equational for the class of all probability functions on any Boolean-closed domain D that contains three logically possible, logically incompatible propositions.

- This follows from the Lemma, since given such a D and appropriate propositions A, B,C, we can construct Pr such that $0<Pr(A)<Pr(A \lor B)<1$.
- But this result is quite boring and obvious, since it is easy to show that for any such D, there is a probability function Pr whose range is {0, 1/3, 2/3, 1} and which assigns 1/3 to two logically incompatible propositions. But no ⇒ can be equational for such a Pr, since if $Pr(A) = Pr(B) = 1/3$ and A and B are logically incompatible, $Pr(A \mid A \lor B) = 1/2$, so $Pr(A \Rightarrow B)$ would have to be 1/2 if any operation ⇒ were equational for Pr.

Second triviality theorem: no binary operation ⇒ is equational for any class of probability functions on the same domain that is closed under conditionalisation and contains some nontrivial probability function (one such that $0<Pr(A \land B)<Pr(B)<1$ for some A, B).

- This follows from the Lemma is much weaker than it and less significant. Not so plausible that rational credence functions/ possible chance functions are closed under conditionalisation on just any old proposition!

Third triviality theorem: no binary operation ⇒ is equational for any class of probability functions on the same domain that is closed under conditionalisation on the members of some finite partition.

8. What about giving up on conditionalisation?

Say that a probability function is *regular* iff whenever $P(A)=1$, A is logically necessary. Some hold that all rational credence functions are regular (on 'anti-Cartesian' grounds). This provides independent reason to doubt that any rational credence function comes from any other by conditionalisation.

Also: in the real world it seems that the complete truth about any interval of history is always so fully specific that it can't have any chance higher than 0. This provides indepedent reason to doubt that the chance function at any time comes from the chance function at any other time by conditionalisation.

However there are strengthenings of Lewis's results that suggest that the problem isn't just with the particular operation of conditionalisation. For example:

Hall's triviality result: if ⇒ is equational for two distinct, non-trivial probability functions Pr and Pr′, then Pr and Pr′ are "near-orthogonal", in the following sense: for every $\varepsilon>0$, there is a proposition A such that $Pr(A)<\varepsilon$ and $Pr'(\varepsilon)>1-\varepsilon$.

It's pretty hard to believe that rational changes of mind always involve moving from one credence functions to another that is near-orthogonal to it!